AAKRITI JAIN

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Google Scholar: scholar.google.com/citations?user=qqWOQgIAAAAJ&hl=en

EDUCATION

PhD, Computer Science *(GPA: 4.00/4.00)* Expected May 2027
University of Texas at Dallas, USA
Advisor: Prof. Abusayeed Saifullah

MSc, Computer Science *(GPA: 4.00/4.00)* 2024
Wayne State University, USA

B.Tech., Electronics and Communications Engineering *(CGPA: 9.3/10)* 2021
ADGITM, Guru Gobind Singh Indraprastha University, New Delhi, India

RESEARCH INTERESTS

Real-Time Systems • Internet of Things (IoT) • Cyber-Physical Systems (CPS) • Wireless Networking • Protocol Design for Low-Power Wide-Area Networks (LPWANs) • Energy-Efficient Embedded Systems • Cross-Technology and Direct-to-Satellite Communications • 5G/6G Networks • Machine-Learning-Based Scheduling

Research Summary. I build low-power wireless systems that enable timely, reliable, and energy-efficient communication for wide-area industrial IoT and CPS, from oil fields and smart grids to remote process plants. In LPWANs, where thousands of energy-constrained devices contend for the same channel, meeting stringent deadlines within tight energy budgets is a fundamental challenge that short-range industrial solutions cannot address. My research brings real-time operation to LPWANs through deadline-aware medium access control, burst-resilient and energy-aware communication for intermittently powered devices, and scheduling–control co-design that treats the network as part of the control loop rather than a fixed delay. My broader work encompasses latency-aware data collection using mobile sinks, cross-technology communication and collision avoidance for Long Range Frequency Hopping Spread Spectrum (LR-FHSS), and machine learning (ML)-based scheduling for multicore processors. I validate these designs on software-defined radio (SDR) prototypes with GNU Radio and USRP B200, on COTS LoRa hardware and multi-node testbeds, and at scale in NS-3, PyTorch, and Gurobi. Going forward, I aim to extend these guarantees to direct-to-satellite and 5G/6G IoT.

CONFERENCE PUBLICATIONS

Each paper's title links to its full text.

[C7] M. A. Haque, **A. Jain**, P. Modekurthy, A. Saifullah. *Enabling Cross Technology Communication from LR-FHSS to LoRa.* In Proc. ACM/IEEE International Conference on Embedded Artificial Intelligence and Sensing Systems (SenSys 2026). CPS-IoT Week 2026. Acceptance rate: 19.8%.

[C6] A. Saifullah, N. Ahmed, **A. Jain**, M. Y. S. Uddin, M. A. Rahman. *Latency-Optimized Data Harvesting in LoRa Networks Using a Mobile Sink.* In Proc. IEEE Real-Time and Embedded Technology and Applications Symposium (RTAS 2026). Acceptance rate: 25.2%.

[C5] **A. Jain**, V. P. Modekurthy, A. Saifullah, M. Brocanelli. *Co-Scheduling of Charging and Transmission in Supercapacitor Powered IoT.* In Proc. IEEE International Conference on Distributed Computing in Smart Systems and the Internet of Things (DCOSS-IoT 2026). Acceptance rate: 23.9%.

[C4] **A. Jain**, M. A. Haque, A. Saifullah, H. Zhang. *Burst-MAC: A MAC Protocol for Handling Burst Traffic in LoRa Network.* In Proc. IEEE Real-Time Systems Symposium (RTSS 2024).

[C3] **A. Jain**, P. Modekurthy, A. Saifullah. *Control over Low-Power Wide-Area Networks.* In Proc. ACM/IEEE International Conference on Cyber-Physical Systems (ICCPS 2024), CPS-IoT Week 2024.

[C2] M. Pivezhandi, **A. Jain**, A. Saifullah, A. Jannesari. *Work-In-Progress: Energy and Thermal-Aware Scheduling based on HMARL for OpenMP DAG Workloads.* In Proc. IEEE Real-Time Systems Symposium (RTSS 2024), Brief Presentations Track.

[C1] S. Fahmida, V. P. Modekurthy, D. Ismail, **A. Jain**, A. Saifullah. *Real-Time Communication over LoRa Networks.* In Proc. ACM/IEEE International Conference on Internet-of-Things Design and Implementation (IoTDI 2022), CPS-IoT Week 2022.

JOURNAL PUBLICATIONS

[J2] **A. Jain**, V. P. Modekurthy, A. Saifullah. *Real-Time Scheduling and Control Co-Design over LPWAN.* ACM

Transactions on Cyber-Physical Systems (TCPS), 2026.

[J1] S. Fahmida*, **A. Jain***, V. P. Modekurthy*, D. Ismail, A. Saifullah. *RTPL: A Real-Time Communication Protocol for LoRa Network*. ACM Transactions on Embedded Computing Systems (TECS), 2024. (* Equal contribution)

RESEARCH FUNDING

- My doctoral research has been supported by seven National Science Foundation (**NSF**) awards (CAREER-2306486, CNS-2301757, CNS-2306745, CNS-2211640, CNS-2601685, CNS-2602744, and CNS-1948365) and one Office of Naval Research (**ONR**) award (N00014-23-1-2151), through collaborative research with faculty and researchers across more than eight universities. 2022 to present

SELECTED TALKS AND POSTERS

Guest Lecture on Traffic-Aware Protocols for Smart Wireless Networks Jan 2025
Wayne State University, Detroit, MI, USA

Burst-MAC: A MAC Protocol for Handling Burst Traffic in LoRa Network Dec 2024
IEEE Real-Time Systems Symposium (RTSS 2024), York, United Kingdom

Poster: Energy- and Thermal-Aware Scheduling Based on HMARL for OpenMP DAG Workloads Dec 2024
IEEE Real-Time Systems Symposium (RTSS 2024), York, United Kingdom

Control over Low-Power Wide-Area Networks May 2024
ACM/IEEE ICCPS 2024, Hong Kong, China

RESEARCH AND PROFESSIONAL EXPERIENCE

Graduate Research Assistant, Dept. of Computer Science Aug 2021 – Present
Wayne State University, USA (Fall 2021 – Summer 2025) and The University of Texas at Dallas, USA (Fall 2025 and Summer 2026) | Advisor: Prof. Abusayeed Saifullah

- Published 7 conference papers and 2 journal papers, with 2 additional manuscripts under review at top-tier conferences.
- Designed real-time MAC protocols for resource-constrained LPWANs. Burst-MAC coordinates bursty LoRa traffic to reduce collisions during time-critical events, while RTPL exploits parallel gateway reception to improve deadline satisfaction. Evaluated in NS-3 and on physical testbeds. *RTSS 2024 (first author); ACM TECS 2024 (co-primary author) and IoTDI 2022.*
- Developed real-time scheduling and control co-design for industrial CPS over LPWANs, combining schedulability analysis and adaptive sensor sampling to optimize control performance under deadlines. *ICCPS 2024 and ACM TCPS 2026 (co-first author)*
- Developed energy-aware charging–transmission scheduling for supercapacitor-powered IoT devices using integer linear programming and online heuristics, and latency-aware mobile-sink data harvesting with a 2-approximation guarantee. *DCOSS-IoT 2026 (first author); RTAS 2026.*
- Built cross-technology communication from LR-FHSS to commodity LoRa receivers and developed per-message delivery certificates and signature-bound frame recovery for direct-to-satellite LR-FHSS using GNU Radio and USRP. *SenSys 2026.*
- Contributed to hierarchical multi-agent reinforcement learning for energy- and thermal-aware scheduling of parallel workloads on multicore platforms. *RTSS Brief Presentation 2024.*

Intelligent Transportation Systems Intern, Delhi Metro Rail Corporation Jun 2019 – Aug 2019
New Delhi, India

- Developed a CNN-based passenger-classification module into the Delhi Metro station-entry surveillance pipeline. Analyzed smart-card transactions using clustering and sequence mining to identify passenger travel patterns for transit planning. Gained hands-on experience with RFID, NFC, and EMI subsystems for automated fare collection.

Undergraduate Research Assistant, ADGITM, GGSIP University Sep 2018 – Feb 2019
Delhi, India

- Led a four-person team building a smart IoT water-level monitoring and control system with a custom multilevel resistive sensor, touch-switch inlet control, and a mobile app for remote monitoring.

Instrumentation and Data Analytics Intern, Indraprastha Gas Limited Jun 2018 – Aug 2018
New Delhi, India

- Developed Random Forest and XGBoost models to detect flow-meter faults caused by sensor aging, environmental drift, and calibration errors, improving detection accuracy by 17% over threshold-based checks. Delivered a predictive-maintenance interface to help field teams identify instrumentation faults.

TEACHING EXPERIENCE

Graduate Teaching Assistant, Dept. of Computer Science Spring and Fall 2026
The University of Texas at Dallas, USA
Course: CS 4390 Computer Networks

- Graded assignments, exams, and project deliverables for both courses.
- Helped students understand course concepts through office hours, question answering, and discussion of networking and systems topics.
- Supported course projects by clarifying requirements, troubleshooting implementations, and providing feedback on design and debugging.

Guest Lecturer, Dept. of Computer Science 2025
The University of Texas at Dallas, USA
Course: Internet of Things (graduate). Delivered multiple lectures on LoRa networking and network simulation in NS-3.

Undergraduate Teaching Assistant, Dept. of Electronics and Communication 2018 – 2019
ADGITM, Guru Gobind Singh Indraprastha University, New Delhi, India
Courses: ETEC 405 Wireless Networks, ETEC 310 Data Communication and Networks, ETEC 305 Microprocessors and Microcontrollers, ETCS 260 Computer Organization and Architecture.

SUPERVISION AND LAB LEADERSHIP

Nasif Ahmed, PhD student, The University of Texas at Dallas January 2025 to present
Junior doctoral student in our laboratory. I mentor him on identifying new problems and approaching their solutions, on experiments with COTS LoRa devices, and on writing systems papers.

Punith Jayaramu, M.S. student, Wayne State University Fall 2024
Guided his transition from foundational wireless communication concepts to independent research, co-developed a Python-based LR-FHSS network simulator, and advised on M.S. thesis writing.

Arefa Haque, M.S. student, Wayne State University Summer 2024
Guided her in implementing LR-FHSS on COTS devices and co-developed a GNU Radio gateway from scratch using a USRP B200, training her in experimental design, hardware debugging, and end-to-end system validation.

AWARDS AND HONORS

Michael E. Conrad Graduate Research Award (Runner-Up). 2024
Wayne State University

Dean's List Award, top 2% of graduating class. 2024
College of Engineering, Wayne State University

Graduate Student Professional Travel Award (two-time recipient). 2024, 2023
Wayne State University

Prime Minister's Extraordinary Academic Achiever Award, top 1% of graduating students nationwide. 2015
Senior Secondary Examination, India

PROFESSIONAL SERVICE

Journal Reviewer

- ACM Transactions on Cyber-Physical Systems (TCPS) 2025 – 2026
- ACM Transactions on Embedded Computing Systems (TECS) 2026
- Pervasive and Mobile Computing (Elsevier) 2023 – 2025
- ACM Transactions on Sensor Networks (TOSN) 2024

Conference reviewer

- IEEE Intl. Conf. on Distributed Computing in Smart Systems and the IoT (DCOSS-IoT) 2026
- IEEE Intl. Conf. on Sensing, Communication, and Networking (SECON) 2026
- International Conference on Networking, Systems and Security (NSysS) 2025
- IEEE Intl. Conf. on Embedded and Real-Time Computing Systems and Applications (RTCSA) 2024, 2025

Conference Referee

- ACM/IEEE Intl. Conf. on Embedded Artificial Intelligence and Sensing Systems (SenSys) 2024 – 2027
- IEEE Real-Time Systems Symposium (RTSS) 2023 – 2026
- Intl. Conf. on Embedded Wireless Systems and Networks (EWSN) 2023 – 2025
- IEEE Intl. Conf. on Distributed Computing in Smart Systems and the IoT (DCOSS-IoT) 2023 – 2025
- IEEE Real-Time and Embedded Technology and Applications Symposium (RTAS) 2022 – 2023

TECHNICAL SKILLS

Programming.	Python, C, C++, MATLAB, SQL, R, Java, Bash
Wireless and SDR:	GNU Radio, USRP B200, LoRa and LR-FHSS physical layers, LoRaWAN, COTS LoRa nodes and gateways
Learning and simulation:	PyTorch, Pandas, NS-3
Systems and tools.	Gurobi, Linux, Docker, Git, LaTeX

REFERENCES

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Dr. Marco Brocanelli
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The Ohio State University
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